# Aluminium-based batteries surpass lithium-ion

*Graduate student researcher Yuhgene Liu holds an aluminium material for solid-state batteries (Credit: Georgia Institute of Technology)*

Researchers at the Georgia Institute of Technology use aluminium foil, presenting a more affordable and eco-friendlier substitute for lithium-ion batteries. The research highlights their potential to incorporate high-performance materials, particularly aluminium, known for its energy efficiency, affordability, and production advantages in the anode, where lithium resides. To combat rapid degradation in pure aluminium foils, researchers integrated trace amounts of diverse materials, forming foils with unique 'microstructures.' Testing over 100 materials yielded improved efficiency and durability compared to standard lithium-ion batteries. The aluminium anode's superior lithium storage capacity results in high energy density, potentially surpassing lithium-ion batteries, paving the way for energy-efficient and cost-effective battery cells. Ongoing work explores aluminium behaviour at larger scales and alternative materials and microstructures for versatile battery configurations, reinvigorating battery research with solid-state technology's promising outcomes.

## Shaping the future of autonomous driving

In a stride forward in the realm of autonomous vehicle technology, the NYU Tandon School of Engineering recently revealed an algorithm named Neurosymbolic Meta-Reinforcement Lookahead Learning (NUMERLA), crafted to surmount the enduring issue of unpredictability and safety in real-world scenarios. In the NUMERLA framework, a self-driving car, when confronted with a dynamic environment, refines its comprehension of the current conditions based on gathered data. Consequently, it anticipates its future performance within a predefined time frame, establishing appropriate safety parameters and refreshing its knowledge base. This approach relies on lookahead optimisation combined with safety considerations, resulting in an empirically secure yet suboptimal online control strategy. By employing this algorithm, the vehicle rapidly adapts to new circumstances without compromising safety. The algorithm's effectiveness was demonstrated through simulations in a digital urban environment, where it outperformed existing algorithms in handling unpredictable scenarios like jaywalkers.

*Autonomous driving (represental image)*